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One World, One Cut: A Real-Wave Interpretation of Quantum Measurement with a Located Boundary

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Contributions. Claude Fable 5.1 (Anthropic; work of 2026-09-01 to 2026-09-05 on this manuscript and its tests, earlier versions under Claude Fable 5 and Claude Opus 4.x as recorded in the repository history) performed the formalization, the manuscript prose, the simulations and their records, and the literature work. John M. Bramble, MD supplied the physical framing and research direction — the real-wave commitment, the register and clock readings, the time–energy–information binding, the temperature dial, the sink formulation of the postulate, and the decision to record the failed mechanism rather than bury it — adjudicated scope and interpretation, and is the accountable and corresponding author. Byline order per standing convention, the accountable author's decision before any circulation. The repository is the record of how every number here was obtained; an archived snapshot with a persistent identifier is to be deposited before circulation, and the commit hash of the reviewed version is given in the review folder.

Abstract. We state an interpretation of quantum measurement and price it. Its commitments are four: the wave is real; there is one world; the Heisenberg cut is a physical crossover with a computable location and width and a physical address, the absorption vertex of a detector site; and outcome selection is a stochastic event at that vertex — the quantum-jump unravelling of photodetection, read as what happens rather than as a calculational device — put in as a postulate. Exact calculations in the smallest models containing the relevant physics are consistent with the following: the crossover in detuning at which irreversible capture sets in has half-width $\sqrt{(\Gamma^2/4 + 2K^2)}$, which is the coupling's population-transfer rate 2K where the coupling

exceeds the record channel's rate Γ, and $\Gamma/2$, set by the record channel and hence by temperature, where it does not — the regime of every room-temperature semiconductor detector; the width is a change of variable and, under the passive-absorber proxy, the inverse quality factor; irreversibility requires a record channel dense on the capture timescale; and neither the carrier frequency nor a self-sustaining amplifier sharpens the crossover. All of this is decoherence theory, and the paper says so. What the paper adds is narrow and stated as such: a location for the event with its regime; the dense-record requirement; the distinction between in-principle recoverability and the reversal an experimenter can apply; and a negative result — that a real-wave, one-world programme with no new constant, forced to reproduce quantum mechanics, is pushed to the projection postulate's branch weights at the decoherence crossover, and that nothing in the physics of a detector, not its record, its amplifier, or its cascade, and not a synchronization substrate, can move the outcome weights. No new constant is introduced and no prediction differs from quantum mechanics.

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1. Scope

Three processes, one dispute. Measurement talk merges three physically distinct processes. (1) *Entangling interaction*: system and apparatus couple; on the joint Hilbert space this is unitary evolution, linear and reversible, and nothing distinguishes a measurement from any other interaction — von Neumann's point, and the origin of the chain. (2) *Decoherence*: the apparatus is open to an environment; the reduced state obeys a master equation that is irreversible for all practical purposes and still linear in the density matrix, so a superposition becomes a mixture of its branches, weighted as before, never one of them. Apparent irreversibility is purchased; selection is not [Schlosshauer 2004]. (3) *The event*: one outcome, in one world. Interpretations disagree about the third. This paper affirms that it occurs, places it at the absorption vertex of a detector site, and postulates it.

What kind of paper this is. An interpretation. It modifies no equation, introduces no constant, and makes no prediction that differs from quantum mechanics as used. In Maudlin's terms [Maudlin 1995] — the wave complete, its evolution linear, measurements with unique outcomes, jointly inconsistent — it gives up linear evolution at the event: it is a collapse account, and §5 states its law. Its claims about the cut (§4) are consistent with exact models run after the claims were first stated in a predecessor paper and before this statement, with two predictions wrong and one definition corrected, all on record; its claim about selection (§5) is a postulate whose form the record's own models forced.

2. Relation to existing frameworks

Decoherence. The cut is the decoherence programme's home ground, and everything in §4 belongs to it. Zeh, Zurek, and Joos and Zeh established that the placement of the split is dynamical: the environment monitors the apparatus in a pointer basis it selects — the preferred-basis result, the programme's central one — and interference between pointer states becomes unobservable at a rate the coupling and the environment's state set [Zeh 1970; Zurek 1981, 1982, 2003; Joos & Zeh 1985]. Joos and Zeh's localization rate and Zurek's decoherence time are coupling-set and temperature-dependent through the environment; so is this paper's crossover (§4.2), and the paper claims no placement decoherence does not already make. Tegmark made the consequence explicit as apparent collapse by scattering [Tegmark 1993]; Brune and colleagues measured a meter's decoherence rate scaling as the square of the separation of its two coherent states [Brune et al. 1996]; quantum Darwinism made the environment's redundant records the criterion of objectivity [Ollivier et al. 2004; Zurek 2009]. The programme's own limit is stated in Schlosshauer's review: decoherence explains the appearance of a cut and not the occurrence of one outcome [Schlosshauer 2004]. Its founders did not stop there: Zeh's reading was Everettian and Zurek's existential interpretation is a hedged position on outcomes, so "decoherence-only" names the results, not the founders. Schlosshauer and Camilleri, asking whether decoherence challenges Bohr's doctrine that a classical side must exist, conclude that it justifies where the cut may be placed without making the cut a physical discontinuity [Schlosshauer & Camilleri 2008; Camilleri & Schlosshauer 2015] — the "for all practical purposes" Bell would not accept [Bell 1990]. This paper accepts every einselection result, including the one that selects the site basis its postulate uses; adds the one thing the programme concedes is missing; concedes that the addition is a postulate; and — v1.1 — concedes that its cut is the decoherence crossover, of finite width, whose in-principle status is exactly what the postulate confers and nothing else (§4.5).

The quantum-jump picture. The formal ancestry of §5 is photodetection theory and the quantum-trajectory unravellings of Markovian dynamics: Glauber's and Kelley and Kleiner's count rate proportional to the local normally ordered intensity [Glauber 1963b; Kelley & Kleiner 1964]; Srinivas and Davies' photon-counting process [Srinivas & Davies 1981]; the Monte Carlo wavefunction and quantum-jump methods of Dalibard, Castin and Mølmer, of Carmichael, and of Plenio and Knight [Dalibard et al. 1992; Carmichael 1993; Plenio & Knight 1998]; and the measurement-theoretic form in Wiseman and Milburn [Wiseman & Milburn 2010]. In that literature a jump is a memoryless event with the golden-rule rate on the conditional state, exactly one jump per quantum for a one-photon state, and a nonlocal update of the conditional state after every jump and every null. This paper's postulate is that unravelling read as what happens, in one world — a position close to a realist reading of Carmichael's, and the paper says what it adds to it (§5.3).

Everett. Denies the event; every branch is real. Whether Everett is local is itself contested [Maudlin 2011; Wallace & Timpson 2010]; the Born weights are supplied by decision-theoretic arguments [Deutsch 1999; Wallace 2012] whose adequacy is contested both ways [Everett 1957 for the relative-state formulation]. This paper makes the opposite trade (§6).

Spontaneous collapse. GRW, CSL, and the Diósi–Penrose models postulate the event as new dynamics [Ghirardi et al. 1986; Bassi et al. 2013; Diósi 1989; Penrose 1996]. GRW and CSL carry new constants; Diósi–Penrose carries none, and its parameter-free form is excluded by the underground test of Donadi and colleagues [Donadi et al. 2021]; CSL's parameter space is constrained by X-ray and non-interferometric bounds and retains allowed regions [Carlesso et al. 2022]. They place the boundary on mass or size; this paper's crossover is coupling- and record-set (§4.2) and it introduces no constant, borrowing the absorber's own coupling — a law with a parameter, but not a new one. Relativistic collapse models exist without a preferred foliation: Tumulka's flash-ontology GRW, Bedingham's and Pearle's relativistic CSL [Tumulka 2006; Bedingham 2011; Pearle 2015]; this matters for §6.

Bohm and Nelson. The family this paper belongs to is the single-world, ψ-ontic, nonlocal one [Harrigan & Spekkens 2010 for the term], and Bohm is its neighbour [Bohm 1952]. Bohm solves the measurement problem with a deterministic guidance law and no event and no cut; this paper postulates an event. Bohm's foliation is empirically inaccessible; so, by this paper's own no-signalling, is this paper's (§6). Nelson's stochastic mechanics [Nelson 1966] is cited for completeness; it faces the Wallstrom objection [Wallstrom 1994] and its author abandoned it over nonlocality, and it is not a close relative.

The transactional interpretation. The nearest neighbour, and v1.1 rewrites the relation against the sources. Cramer's interpretation, built on Wheeler–Feynman absorber electrodynamics and its quantum extensions by Hoyle and Narlikar, Pegg, and Davies [Wheeler & Feynman 1945; Hoyle & Narlikar 1969; Pegg 1975; Davies 1971, 1972], describes a quantum event as a transaction between an emitter's retarded offer wave and an absorber's advanced confirmation wave, and claims that the Born weight follows as their product [Cramer 1986]. Kastner's relativistic and possibilist version adopts Davies' direct-action electrodynamics with fermionic currents as sources [Kastner 2012], and — the point this paper's v1.0 mis-stated as its own — already places the Heisenberg cut at the level of absorption by individual atoms or molecules with the coupling as the physical variable, absorption "quantitatively defined" [Kastner 2018; Kastner & Cramer 2018]; Kastner has also stated its relation to decoherence [Kastner 2020], and Cramer and Mead have given a microscopic two-atom model of transaction formation [Cramer & Mead 2020]. Kastner and Cramer claim to derive the Born rule for radiative processes from the requirement that a real photon needs both an emission and an absorption vertex; that derivation has been disputed [Marchildon 2017; Kastner 2017 reply], and this paper's view — that the identification of the confirmation amplitude with ψ^* stipulates the square — is one side of that dispute, not a settled fact, and is stated as such in §5.3. Two differences of ontology and price. Kastner's offer and confirmation waves are elements of a space of possibilities, not fields on spacetime, so possibilist TI is not a neighbour on this paper's first commitment; Cramer's original is closer. And both Cramer and Kastner claim their nonlocality comes without a preferred frame, the transaction being tied to a spacetime interval rather than an instant; this paper carries a foliation (§6) that TI does not, which is the largest difference of price between the two and cuts against this paper. What this paper adds to

RTI's placement of the cut is narrow: the rate-balance form of the location with its regime, the dense-record requirement, the distinction between the echo and the coupling flip, and the negative result of §5.2.

Curie–Weiss models of registration. Allahverdyan, Balian, and Nieuwenhuizen solve the dynamics of the record's amplification and assume selection at the crucial step [Allahverdyan et al. 2013]. This paper agrees that registration is statistical mechanics and places the event one stage earlier (§4.3) — a placement that, by von Neumann's mobility result, no experiment can contradict, which the paper says in §4.3.

Von Neumann–Dirac, Wigner, the modal interpretations, Copenhagen, QBism. The paper's closest kin in the textbook tradition is objective collapse somewhere in the chain — Bell's "jump" — given a physical address; the modal interpretations (single world, ψ real, a rule for which observable has a value) are nearer relatives than Nelson. Bohr held the cut to be a movable convention and the wave a symbolic device; QBism holds the state to be an agent's belief and the cut the boundary of that agent's knowledge. This paper takes the wave to be real and the cut to have an address, and pays for the realism in §3.6 and §6.

3. Ontology

3.1 Real waves, and on what space. For one quantum the wave is a physical field on spacetime: the Dirac field for matter, the Maxwell field for light. For more than one quantum the state of the field is a functional on configuration or Fock space, and this paper — v1.1, an adjudication the accountable author had reserved and the first author made on his stated recommendation — adopts configuration-space realism explicitly [Albert 1996; Ney & Albert 2013]: the beable is the quantum state of the field, real, on that space. This is a cost, Schrödinger's second objection, and it is paid here rather than held open; §5.1 clause 3 and §6 cannot be stated without it. Two further honesties. The fermionic case is harder than the bosonic: the classical Dirac field is Grassmann-valued, and what a real fermionic field beable is remains unsettled [Struyve 2010]; the paper's calculations use two-level absorbers and never a spinor, so its "Dirac field" is a commitment, not a working part. And the ψ-ontic commitment is supported, under a preparation-independence assumption, by the Pusey–Barrett–Rudolph theorem [Pusey et al. 2012], whose conclusion concerns exactly the configuration-space state adopted here.

"Particle" names a persisting excitation; persistence is a property of bound states — the displaced pole of a dressed excitation — needing no mechanism beyond the equations. The virtual/real distinction is propagating versus evanescent.

3.2 Superposition as coexistence. A superposition is both components present with a definite relative phase — coexistence, never alternation. When the components differ in energy their beat at the Bohr frequency is the laboratory signature that both are there: quantum beats, Ramsey fringes, and Schrödinger's *zitterbewegung* for a Dirac wavepacket containing both energy signs [Schrödinger 1930]. On the Bloch sphere the free dynamics precesses the phase at fixed latitude; the outcome populations are constants of the unitary motion and alternation would destroy the Born weights. The latitude carries the weight; the move from latitude to pole is the event of §5.

3.3 The register. One two-sector structure recurs: the electron's chiral pair, the photon's helicity pair, the interferometer's which-arm pair — an SU(2) register storing its observable as a relative amplitude and phase, intrinsic in the first two cases and geometric in the third, where a beamsplitter sculpts the mode function into two branches. Path superposition is not a puzzle about a delocalized photon; it is the delocalization, shaped [Grangier et al. 1986]. Polarization is the helicity register; the Poincaré sphere is its Bloch sphere. Delocalization is the wave's birthright; discreteness is the detector's budget — one quantum, one closure, however thinly the amplitude is spread. Which-path is the register electromagnetic detectors einselect, because the interaction couples through the vector current; that is a decoherence result and the paper uses it as one.

3.4 The photon. Vacuum Maxwell in Riemann–Silberstein form splits into two decoupled helicity sectors [Bialynicki-Birula 1996]: the register with the off-diagonal coupling set to zero, no mass, no beat, no internal clock. Matter can lend the photon a coupling; everything clock-like about light is borrowed. A massive excitation carries a real phase at the Compton frequency, demonstrated interferometrically [Lan et al. 2013]; it is a phase, representation-dependent [Foldy & Wouthuysen 1950], and not a coupling.

3.5 Two classical limits. Matter becomes classical by being recorded, one absorption at a time [Tonomura et al. 1989]; Pauli exclusion bars a fermion field from macroscopic occupation of one mode, so there is no classical electron wave, and matter reaches the occupancy route only by first becoming bosonic. Light becomes classical by occupancy, the coherent state [Glauber 1963a]; thermal light is classical only as noise [Hanbury Brown & Twiss 1956]. A single photon, too free to be recorded in flight and too small to be loud, is excluded from both, and at interaction is annihilated, converting itself into a record in matter.

3.6 One world. Exactly one outcome occurs. Bell's theorem, given settings independence and no conspiracy, presents the bill: a single world with definite outcomes that reproduces the quantum correlations is nonlocal [Bell 1964]. The trade is made in §6.

4. The cut, with an address and a regime

Each claim is cited to the exact calculation that bears on it (`heisenberg_cut_recoverability/` in the repository, including `REVIEW_RUNS_RESULTS.md` for the v1.1 additions). The calculations are textbook — Rabi occupation, Weisskopf–Wigner decay, the Bloch–Siegert shift, Poincaré recurrence of a discretized bath, the forced Stuart–Landau oscillator — and are presented as what the interpretation's placement is consistent with, not as new physics.

4.1 The criterion, and a disclosure. The cut is where *in-principle recoverability ends*: the first point in the chain at which no operation on the site and the field, however idealized, restores the pre-event configuration. Operationally, recoverability is the partial Loschmidt echo — the site's own Hamiltonian reversed for the capture time, the environment untouched. This is the irreversibility of the reduced dynamics, which is what decoherence theory means by irreversibility; the paper's v1.0 claimed more (§4.5). The echo is not the reversal an experimenter can apply to the site–field coupling alone: even with no environment, flipping the coupling returns a detuned capture only on resonance, because the detuning keeps running forward; the echo returns it exactly, in the model, at every detuning. The disclosure: the record's first definition *was* the coupling flip, and it failed the no-environment calibration (a return of 0.04 off resonance); the echo definition was adopted after that run and before the results were opened. Experiments that locate the cut by a reversal protocol must separate the two.

4.2 Location, regime, and width. A site holding energy $e < E_0$ is off-shell with deficit ΔE, and the deficit sets a return rate, κ_ret $= \Delta E/\hbar$ — a Wigner–Weisskopf-era heuristic for the rate at which a virtual excitation returns to the field, which in the linear model is the detuning. Irreversible capture sets in over a crossover in detuning whose half-width, in the smallest model with capture, return, and a dense record channel of rate Γ, is

$$\Delta_{1/2} = \sqrt{\Gamma^2/4 + 2K^2},$$

verified across Γ/K from 0.2 to 1000 (Run A: midpoints 1.54, 1.28, 5.09, 50.0, 500.0 against the formula's 1.42, 1.50, 5.20, 50.0, 500.0). Two regimes. For $\mathbf{\Gamma \ll K}$ the half-width is 2K, the coupling's population-transfer rate — the Rabi half-width of a power-broadened line — and the crossover is coupling-set; the rate-based midpoint in the record's 256-mode model is κ_ret/K = 1.3–2.0 across two decades of Γt. This is the regime of rare-earth memories and cavity QED at strong coupling. For $\mathbf{\Gamma \gg K}$ the half-width is $\Gamma/2$: the crossover is set by the record channel's rate, which is the temperature-dependent quantity, and every room-temperature semiconductor detector, at Γ/K of order 10^5, sits here. The v1.0 claim that the crossing is "coupling-set rather than temperature-set" was true of the first regime only and is withdrawn as a distinctive claim; in both regimes the placement is decoherence theory's own. The midpoint read from recoverability at a fixed observation time drifts outward with Γt (from 1.7 to 5 across Γt = 0.15 to 10), so the location is a statement about rates and the crossover has relative width of order one; a boundary of that kind is a scale, which is the sense in which Schlosshauer and Camilleri are right that the cut is not a discontinuity.

The width. The layer within which a site's restoring rate is below the coupling has share width $w = K/\omega$, and this is a change of variable from the crossover above: with share $s = e/E_0$ and $\Delta E = (1 - s)\hbar\omega$, κ_ret/K ~ 1 is $1 - s \sim K/\omega$; the carrier ω cannot appear in a rotating-wave model and did not, and enters only as the Bloch–Siegert shift of the crossover's centre, $-(1.1–1.7)K^2/\omega$, with the half-width unchanged to 2 % for

$\omega/K \geq 8$ [Bloch & Siegert 1940]. Under the passive-absorber proxy K ~ Γ — defensible where the coupling to the field is the interaction that sets the linewidth, dropped where K is engineered independently — the width is Γ/ω, the inverse quality factor of the transition, and it is tabulable for discrete-transition absorbers: about 10^{-8} for an alkali line, 10^{-18} for an optical clock transition, 10^{-7}–10^{-6} for a transmon, 10^{-6}–10^{-5} for a quantum dot [Steck 2025; Ludlow et al. 2015; Kjaergaard et al. 2020; Kuhlmann et al. 2015]. The table says that high-Q two-level systems are clean two-level systems, which is Leggett's disconnectivity in other words and dissolves no puzzle on its own. A band absorber — silicon, a nanowire, a calorimeter — has no single transition for a site to be detuned from; for it neither Δ nor a "site" is given by the Hamiltonian, the site basis is einselected by the local phonon coupling, and the location claim has no referent beyond the $\Gamma/2$ regime statement above.

exact-model result	value	status
crossover half-width vs Γ/K (Run A)	$\sqrt{(\Gamma^2/4 + 2K^2)}$ to 15 % from Γ/K = 0.2 to 1000	textbook (adiabatic elimination); fixes the regime
rate-based midpoint, Γ ≪ K, 256-mode channel	κ_ret/K = 1.3–2.0 over Γt = 0.15–10	the Rabi half-width
carrier frequency, rotating-wave approximation removed	centre $-(1.1\text{–}1.7)K^2/\omega$; half-width within 2 % for $\omega/K \geq 8$	Bloch–Siegert
record channel of 1, 4, 16, 64, 256 modes	coherent exchange; nothing (modes off resonance); recurrence; 64 agrees with 256 to 0.005 for t ≤ 3	Jaynes–Cummings to Weisskopf–Wigner
self-sustaining amplifier swept through its gain (classical)	stored energy at the field-set value through both Hopf points (slope 0.004), rising past the running onset (slope 1.1)	synchronization theory; the amplifier kinks, does not sharpen

4.3 Three stages, one crossing.

stage	dynamics	recoverability
capture	unitary, distributed across candidate sites in proportion to local intensity	recoverable — demonstrated for controlled ensembles by spin echoes, photon echoes, and atomic-frequency-comb memories [Hahn 1950; Kurnit et al. 1964; de Riedmatten et al. 2008; Afzelius et al. 2009]; the coherent, reversible onset of a monitored quantum jump [Minev et al. 2019] is suggestive of the same for a single system
commitment	the absorption vertex and thermalization at one site: for a semiconductor the interband vertex, of order 10 fs–1 ps (literature-typical, spanning two orders)	**the event; recoverability ends here by postulate**

stage	dynamics	recoverability
registration	reservoir-powered amplification	irrecoverable and macroscopic; multiplies the vertex weight by a stochastic efficiency that depends on position and energy and adds photon-independent clicks; contributes no interference-sensitive weight

Commitment is the vertex, not the record. The v1.0 row "photon-agnostic, no which-site weight" was wrong as written: a nanowire's internal efficiency is a stochastic function of bias current, photon energy and absorption position [Renema et al. 2014; Engel et al. 2015], an avalanche diode's breakdown probability depends on where the carrier is generated, afterpulsing and crosstalk give clicks with no vertex, and in a calorimeter thermalization is the registration. What registration does not carry is *interference-sensitive* weight: a POVM efficiency η(x, E) multiplies the vertex's branch weight and cannot move the relative phases the vertex has already discarded. And by von Neumann's mobility result, moving the event from the vertex to the amplifier within a decohered chain changes no prediction; "the record is written later" is a statement the paper makes about its own ontology and the world cannot contradict. Two physical requirements: the record channel must be dense on the capture timescale (one record degree of freedom gives coherent exchange, not irreversibility); and the amplifier's engaged regime — a free phase exists but is entrained — leaves the stored energy where the field set it, the bookkeeping changing hands from field to site only at the onset of free running, continuously.

4.4 The detector taxonomy. The photon-counting detector is the projective limit: one quantum, one vertex, one click at saturated internal efficiency, the only class for which per-event Born claims are properly made. NMR/MRI is the ensemble limit. The cloud chamber is the repeated limit, a sequence of capture–commit–register triplets [Mott 1929]. The photographic emulsion is the integrating limit [Gurney & Mott 1938]; that its reciprocity failure — a latent speck decaying if not reinforced — evidences a commitment/registration distinction is this paper's inference, not Gurney and Mott's. Every click deposits the full quantum at one site; distributed capture is reversible energy, never banked; losing sites retain no residue.

4.5 What recoverability is, and what its ending means here (v1.1: the in-principle claim conceded). A reversal requires the stored phase history to point back along the path. Whether it survives is set by the reference structure of the medium: references *common* across sites give an invertible record; references *different but deterministically known* give a record invertible by rephasing, which is why echoes work; references *re-randomized* on the dephasing timescale by a thermal bulk give a record that is not invertible by any operation on the site and the field. Heat is the writer: thermal motion delivers phase kicks at random times, the record of the incident wave's history spreads over e^{S/k_B} configurations of the medium, and reversing it requires acting on the medium. The v1.0 of this paper claimed that this ends recoverability *in principle* rather than for all practical purposes. Three reviewers showed independently that it does not: the environment is excluded from the reversal by the criterion's own definition; a thermodynamic price is not a prohibition, and Landauer's bound prices erasure, not reversal, which Bennett showed carries no such minimum for an agent holding the microstate [Bennett 1973, 1982]; the polarization-echo results cited for the strong form [Levstein et al. 1998; Pastawski et al. 2000] concern how imperfect reversals fail and presuppose imperfection; and a Poincaré recurrence is not forbidden. The claim is conceded. Recoverability of the reduced dynamics ends for all practical purposes, as decoherence says; what ends it in principle is the event of §5, which is postulated, and nothing else.

5. Selection at absorption

5.1 The postulate, in the language of a sink (v1.1: clause 2 restated). A quantum arrives at a detector as a wave spread over many sites. Three things then happen, and the paper postulates all three.

1. *Every site draws, in proportion to the square.* A detector site at the single-photon level is a linear

absorber: it has no phase of its own, the wave creates its oscillation, and the phase difference between the two is fixed by the site's detuning and return rate. The energy the site draws from the field is proportional to the square of the local amplitude, weighted in its detuning by the crossover of §4.2. Eligibility is graded, not sharp; no site is excluded; a background rate, thermal or current-assisted, exists at every site and gives the dark counts a hazard on drawn energy alone would lack.

2. *A sink opens at one site, by chance, on the conditional state.* At each site a sink can open at any moment, with a probability per unit time equal to the golden-rule rate times that site's **branch weight in the conditional state** — the state of the whole quantum renormalized by every null that has occurred elsewhere, as in the quantum-jump unravelling [Srinivas & Davies 1981; Dalibard et al. 1992; Carmichael 1993; Wiseman & Milburn 2010]. The first sink to open anywhere is the one that opens. This is *not* a hazard on the whole quantum's survival that is linear in the energy the site has drawn, which is what v1.0 said: the record's own model shows that rule sending the click to whichever detector the wave reaches first whenever the arms are unequal — every Bell test — and making a detector's click probability an exponential of intensity rather than linear (Run B: with sequential exposure the nearer channel wins in every trial in which it fires, and a channel at half intensity clicks 36 % more often than a linear law allows). The rule that reproduces quantum mechanics is the branch weight, and the null-conditioned renormalization that keeps it right is itself nonlocal. *By chance*: not when the site's phase reaches alignment, which makes the winner the fastest of N nearly deterministic approaches with the wrong statistics.

3. *The whole quantum flows through it.* Once a sink is open, the quantum's entire energy passes through that one site, and the many-quantum wavefunction is projected onto that event: for a one-photon state every other amplitude of that photon vanishes — including in the other arm of an interferometer, metres away — within a time the record's model bounds at a thousandth of the opening time, orders of magnitude faster than light crosses the arms; for a two-photon state, the projection acts on the two-photon amplitude, which is why the herald's click leaves the signal photon intact and why two indistinguishable photons at a beamsplitter never open sinks at both outputs while two distinguishable ones do half the time. The projection is on Fock space (§3.1); a field on three-space cannot carry it. One sink per quantum: for a one-photon state exclusivity is kinematic, since the two-count correlation of the field vanishes, and the paper's "stop" is the projection, not a dynamical race.

Stated formally. The matter–field state evolves by the stochastic master equation of a photodetection unravelling: between jumps, the no-jump Hamiltonian H − (iħ/2)Σ_i Γ_i P_i with P_i the site's absorbing projector, renormalized; at rate Γ_i $\langle\psi|$P_i$|\psi\rangle$ a jump onto site i's absorbed state. Its ensemble average is a Lindblad equation, linear in ρ, and therefore does not signal [Gisin 1989; Polchinski 1991] — no-signalling is a theorem of the adopted formalism, not a consequence of hazard linearity as v1.0 claimed. Clause 1 is the golden rule's matrix element squared; clause 2 is the jump; clause 3 is the projection with Born weights, together with the nonlocality that one world costs (§6). The paper derives none of the three, and they are the projection postulate placed at the vertex.

5.2 What the record established, and a negative result. The postulate's form was not chosen freely. A programme to derive it from a synchronization dynamics in a real-wave substrate — a race among detector-site oscillators locked to the field — was tested before this paper was written and is recorded as a negative result (`NEGATIVE_RESULT.md`; `adler_two_channel_exploratory/RESULTS.md`; every race number below carries the package's own `diagnostic_only` label and none entered a production estimate). What the record established is this. A first-to-align rule among self-sustaining oscillators gives an outcome exponent of 1.5 in the coupling with the site count entering as its logarithm — a fact about phase-oscillator substrates. A memoryless hazard linear in absorbed energy on such a substrate gives 2.1–2.2, the Born value plus a tenth of a power that persists across grid, window, pulse, noise, and commitment time and whose mechanism the record calls consistent with an extreme-value ingredient and not established. A graded per-site weight proportional to the squared amplitude gives Born exactly, by the competing-exponentials identity — which is the Born rule assumed in the hazard, and is why synchronization was dropped. A hazard on local drawn energy fails for staggered exposure (Run B). And nothing downstream of the vertex — the record, the amplifier, the cascade — carries interference-sensitive weight (§4.3). Put together: **a real-wave, one-world programme with no new constant, forced to reproduce quantum mechanics, is pushed to the**

projection postulate's branch weights at the decoherence crossover; nothing in the physics of a detector and no synchronization substrate can move the outcome weights, and the family of "hazard at the absorber" mechanisms collapses onto the projection postulate. That is this paper's one result, and it is a no-go.

5.3 Relation to the transactional picture, exactly. Both put actualization at the absorber. The transactional interpretation claims to derive the square from the requirement that a real photon has both an emission and an absorption vertex; this paper reads the identification of the confirmation amplitude with ψ^* as a stipulation of the square, which is Marchildon's side of an open dispute [Marchildon 2017; Kastner 2017], and postulates the square outright. Both are nonlocal; TI claims to be so without a foliation and this paper is not (§6). What this paper adds to RTI's placement of the cut at atomic absorption [Kastner & Cramer 2018] is the rate-balance location with its regime, the dense-record requirement, the echo/flip distinction, and the no-go of §5.2. Neither predicts anything quantum mechanics does not.

6. Entanglement and the price of one world

Two objects. The local thermal reference at each detector — the re-randomized references that make commitment possible — is established in the common past of both wings and is independent of the analyzer settings: it is a Bell common cause, and by Bell's theorem it cannot source the correlations [Bell 1964]; a clock-as-local-variable model of the outcomes is sub-classical, CHSH $\leq \sqrt{2}$ in the record. The correlation comes from the second object: the two-particle state on configuration space (§3.1), non-separable, spanning both wings. When one wing's sink opens, the partner wing's state is updated to the conditional state — the Lüders rule, which is part of the postulated jump (§5.1 clause 3), not a restatement of anything. Settings are free; no-signalling is the theorem of §5.1.

The trade. Everett buys locality, if it has it, with many worlds; this paper buys one world with nonlocality in Bell's sense, never signalling. No interpretation that accepts Bell's premises and unique outcomes pays neither price; retrocausal, superdeterministic, and QBist accounts decline the premises.

The foliation, as a choice. The jumps of §5.1 occur at events, and for spacelike-separated jumps on one state a global ordering is needed to say which conditional state the second jump acts on. This paper orders them by a preferred slicing. That is a choice with alternatives: Tumulka's flash ontology orders collapses covariantly, Bedingham's and Pearle's relativistic CSL do without a foliation, and Dürr and colleagues extract a foliation covariantly from the state itself [Tumulka 2006; Bedingham 2011; Pearle 2015; Dürr et al. 2014; see also Aharonov & Albert 1984; Myrvold 2002]. The paper does not argue that its choice is forced; it records it, with its cost. The slicing enters wherever a jump occurs, which is at every absorption with a dense record channel and not only in a laboratory — v1.0's "confined to measurement" is withdrawn — and as written it is a low-energy Lorentz violation whose clearance of every established bound is owed; whether the frame is a vacuum rest frame (Lorentz violation under the full weight of the clock-comparison constraints) or a matter or radiation rest frame (constraints largely inapplicable) is a fork the programme's record holds open. By the paper's own no-signalling the foliation is unobservable in any correlation, exactly as Bohm's is; v1.0's claim that it "carries tests" is withdrawn (§7).

7. Consequences and tests

No experiment separates this interpretation from quantum mechanics with the projection postulate, because it is that. What follows is an honest inventory.

Tests the interpretation shares with decoherence theory, and passes because quantum mechanics does. The crossover of §4.2, in its coupling-set regime, would show in a strongly coupled cavity-QED or rare-earth system as a detuning crossover of half-width 2K in the echo-recoverable fraction, and in its record-set regime as $\Gamma/2$ — consistency tests of textbook physics. Retrieval efficiency versus storage time at several temperatures collapses in $\Gamma(T)\cdot t$ for a memoryless channel, which is the definition of T_2, with Mims spectral-diffusion decay as the known non-collapse branch [Böttger et al. 2006]. The circuit-QED coupling dial — commitment latencies and the echo-recoverable window as the engineered coupling ratio is swept — has in substance been done [Katz et al. 2008; Murch et al. 2013; Hatridge et al. 2013], gives the master-equation curve, and vertex latencies of 10 fs are inaccessible to any readout [Korzh et al. 2020]. Against the mass-based collapse models the dial discriminates only in principle, since at bounds-consistent parameters their added rate lies far below environmental dephasing.

The frame tests, withdrawn. v1.0 offered sidereal and before-before tests of a cosmic frame. Since the interpretation does not signal, the foliation modulates no correlation, and a null retires nothing; the before-before and 24-hour experiments exist and are null [Zbinden et al. 2001; Stefanov et al. 2002; Salart et al. 2008]. A positive sidereal signal would be a departure from quantum mechanics that this interpretation does not predict. The programme's one worked candidate for a frame signature requires an additional non-covariant coupling that does not follow from the interpretation plus electrodynamics; it is a separate postulate with its own test, recorded in the repository, and not this paper's.

What falsifies what. Whatever falsifies quantum mechanics with the projection postulate falsifies this paper: a single-photon detection efficiency nonlinear in intensity, a two-photon coincidence statistic outside Fock-space projection, a Bell correlation that signals. A boundary located by mass on the coupling dial would falsify the placement of §4.2 and favour a collapse model. Curvature in the conditional statistics of a photon split between two ports of mismatched structure — which any local POVM makes affine in the splitting ratio — would overturn the linear structure of quantum mechanics itself.

8. Open problems

1. **The quantum self-sustaining amplifier.** The engaged and running regimes of §4.3 were shown in a classical oscillator with gain by hand; the open-system calculation with a dense record channel is not done.
2. **The covariant formulation.** Whether the postulate can be stated on a flash ontology or a state-derived foliation rather than a preferred slicing, and the clearance of the Lorentz bounds if it cannot.
3. **The fermionic beable** (§3.1), and the status of configuration-space realism as the paper's ontology.
4. **The residual above the Born exponent** in the synchronization record, consistent with an extreme-value ingredient and not established — a question about phase-oscillator substrates, retained because the record is public.

9. Conclusion

Schrödinger built an equation that describes the evolution of the wave and meant it as physics; the description was extended to relativity and to fields and then stopped, one postulate short. This paper does not remove the postulate, and its one result is that a real-wave, one-world programme that tries to cannot do so from the physics of the detector: the record, the amplifier, the cascade, and a synchronization substrate all leave the outcome weights where the projection postulate puts them. What the paper gives the postulate is an address with a regime, a criterion that separates in-principle recoverability from the reversal an experimenter can apply, a requirement that the record be dense, and a price list: a real state on configuration space, a nonlocal projection, and a foliation chosen rather than forced. The wave is real and evolves; where it is absorbed, one site commits, in one world; and what the postulate does not explain is stated as a postulate and priced.

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